

Experiment 3

Aim: Exploratory Data Analysis & Statistical Analysis

- Objective:**
1. To visualize the distribution of classes and features in the dataset.
 2. To understand the spread and central tendency of data using plots.
 3. To identify correlations between features using heatmaps.
 4. To perform statistical hypothesis testing (e.g., t-tests) to determine if observed differences are significant (as per the requirement).

Detailed Steps

- **Plot class balance:** Use a count plot or pie chart to visualize the number of samples in each class.
- **Visualize frequency distributions of features:** Use Histograms and boxplots
- **Assess Feature Correlations :** Use Heatmap for correlation matrix
- **Feature Distribution Analysis & Outlier Detection:** fit a distribution (e.g., Gaussian, Poisson etc) to continuous/discrete variables.
- **Hypothesis Testing :** Run t-test/chi square/ANOVA for mean differences
- **Document EDA insights**

Open-Source Tools

Matplotlib, Seaborn, Plotly, SciPy, Statsmodels

Deliverables

A well-commented **EDA Notebook** including:

1. Data loading
2. Visualizations (plots, heatmaps)
3. Statistical test implementations
4. Summary of insights

Hypothesis test results, clearly documenting:

1. Hypotheses
- Test statistic
2. P-value and interpretation
3. Conclusion based on test result

Conclusion

